

$$\begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & k & 1 \\ k & 1 & -1 \end{array} \right]$$

$$\downarrow R_2 \rightarrow R_2 - kR_1$$

$$= \left[\begin{array}{cc|c} 1 & k & 1 \\ 0 & 1-k^2 & -1-k \end{array} \right]$$

(a) For ∞ solutions

$$1-k^2=0 \quad \text{and} \quad -1-k=0$$

$$\Rightarrow k = \pm 1 \quad \Rightarrow k = -1$$

$$\Rightarrow k = -1$$

(b) For No solution

$$1-k^2=0 \quad \text{and} \quad -1-k \neq 0$$

$$\therefore k = \pm 1 \quad k \neq -1$$

$$\Rightarrow k = 1$$

(c) For unique solution

$$1 - k^2 \neq 0$$

and

$$1 - k \neq 0$$

$$\therefore k \neq \pm 1$$

and

$$k \neq -1$$

$$\Rightarrow k \neq \pm 1$$